

From Two-Sided Density to a Radial Graph: what the realization step really needs (and what it does not)

Abstract

We isolate the two independent inputs of the “realization” step in the selection-free route to quantitative Saint–Venant: (A) an *inclusion* $B_{r_-}(z) \subset \Omega \subset B_{r_+}(z)$, and (B) the *realization* of Ω as a radial graph with a *quadratic* pocket estimate $|G \setminus \Omega| \lesssim (P(\Omega) - 2\pi)^2$. The first version of this note claimed that (B) holds for any set of finite perimeter. **That claim is false**, and we begin by giving the explicit counterexample (a “lollipop”: a wide cap overhanging a thin stem), which shows that finite perimeter alone never controls the pocket volume quadratically — not even inside an arbitrarily thin annulus, and not even under two-sided density. The correct hypothesis is a purely measure-theoretic *no-overhang* condition, $\mathcal{H}^1(\partial^* G \setminus \partial^* E) = 0$, which is far weaker than $C^{1,\gamma}$ but strictly stronger than finite perimeter. Under no-overhang we prove the perimeter decomposition and the sharp quadratic pocket bound (Theorem 4); this is the honest GMT form of the Plan 4 pocket lemma, and it does retire the $C^{1,\gamma}$ hypothesis *provided no-overhang is available*. We then analyse precisely what input (A) provides: an annulus gives the trivial linear bound $|G \setminus \Omega| \lesssim \eta$ unconditionally, while two-sided density plus tentacle exclusion removes overhangs whose *necks* sit above scale r_* , leaving a genuine residual of sub- r_* overhangs. The upshot is a clean trisection of the programme into inclusion, no-overhang, and the radial-graph theorem, with the residual identified honestly.

1 The two inputs of realization

Fix the planar normalisation $|\Omega| = \pi$, comparison ball $B = B_1$. The goal of the realization step is to replace a (possibly rough) competitor Ω by a *radial graph* G with $|G \triangle \Omega|$ controlled and $\delta_T(G)$ controlled, so that the small-amplitude radial-graph theorem (G) applies. Two logically separate things are needed:

- (A) *Inclusion*. A centre z and radii $r_- < r_+$ with $B_{r_-}(z) \subset \Omega \subset B_{r_+}(z)$.
- (B) *Realization*. From (A), the radial hull G (star-shaped envelope from z) is a radial graph, and $|G \setminus \Omega|$ is controlled *quadratically* by the perimeter deficit.

The historical role of selection was to make $\partial\Omega$ regular enough ($C^{1,\gamma}$) that both (A) (via BNST) and (B) (via the polar-slicing pocket lemma, which used Sard and the implicit function theorem) could run. The question this note addresses is: *how much regularity does (B) actually need?* The answer is more subtle than “none”: it needs a no-overhang condition. We first show, by an explicit example, why finite perimeter is not enough; then we prove the sharp statement under no-overhang; then we examine what (A) supplies.

2 Input (A): inclusion from density, recalled

We recall the mechanism of `almost_serrin_discussion.tex`. Let u be the torsion function of Ω , $c_\Omega = |\Omega|/P(\Omega)$, and define the Serrin and isoperimetric deficits

$$\varepsilon^2 := \int_{\partial\Omega} (|\nabla u| - c_\Omega)^2 d\mathcal{H}^1, \quad \delta_{\text{iso}} := \frac{P(\Omega)}{2\sqrt{\pi|\Omega|}} - 1.$$

The local flux identity (integrate $-\Delta u = 1$ over $\Omega \cap B_r(x)$) combined with Cauchy–Schwarz against ε and the relative isoperimetric inequality in balls yields the *tentacle-exclusion* estimate: for $x \in \partial\Omega$,

$$\varepsilon_r^2 \gtrsim |\Omega \cap B_r(x)|^{1/2} \quad \text{when } |\Omega \cap B_r(x)| \ll r^2,$$

so small Serrin deficit forbids thin interior tentacles above scale $r_* \sim \varepsilon^2$; quantitative isoperimetry forbids exterior fjords above scale $\delta_{\text{iso}}^{1/4}$. The outcome is the two-sided density bound

$$c_0 r^2 \leq |\Omega \cap B_r(x)| \leq (1 - c_0) r^2, \quad x \in \partial\Omega, \quad r \in [r_*, r_0], \quad (1)$$

with $r_* \sim \varepsilon^2 + \delta_{\text{iso}}^{1/4}$, and (via an adapted Magnanini–Poggesi step) the inclusion

$$B_{\rho_i}(z) \subset \Omega \subset B_{\rho_e}(z), \quad \rho_e - \rho_i \leq \eta := C(\varepsilon^\theta + r_*). \quad (2)$$

We treat (1)–(2) as the *output* of input (A); its proof is the content of `almost_serrin_discussion.tex` (with the open points there). What matters below is only that (A) delivers an inclusion with some width η , a finite perimeter $P(\Omega) < \infty$ (immediate from (1), Ahlfors regularity), and the *scale* r_* above which density holds.

3 Why finite perimeter is not enough

We work in $\mathbb{R}^2$ with polar coordinates about 0. For a set E of finite perimeter with $B_{r_-}(0) \subset E \subset B_{r_+}(0)$, define the radial slices, the profile, and the radial hull

$$E_\theta := \{r : re^{i\theta} \in E^{(1)}\}, \quad R(\theta) := \text{ess sup } E_\theta, \quad G := \{re^{i\theta} : 0 \leq r < R(\theta)\}, \quad (3)$$

and the pocket set $Q := G \setminus E$, with connected components $\{Q_j\}_{j \in J}$. The realization step rests on the would-be *perimeter decomposition*

$$P(G) + \sum_{j \in J} P(Q_j) \leq P(E) \quad (\text{false in general}). \quad (4)$$

We exhibit a finite-perimeter E for which (??) fails by a fixed positive amount.

Proposition 1 (Finite perimeter is insufficient). *Let $0 < a < b < R^+$ and $0 < w' < w < 2\pi$, and set (the “lollipop”)*

$$E = B_a \cup \underbrace{\{a \leq r \leq b, |\theta| \leq w'/2\}}_{\text{stem } S} \cup \underbrace{\{b \leq r \leq R^+, |\theta| \leq w/2\}}_{\text{cap } C}.$$

Then E has finite perimeter, $B_a \subset E \subset B_{R^+}$, its radial hull is $G = \{|\theta| < w/2, r < R^+\} \cup \{|\theta| > w/2, r < a\}$ with two pockets $Q^\pm = \{w'/2 < \pm\theta < w/2, a < r < b\}$, and

$$P(G) + P(Q^+) + P(Q^-) - P(E) = 4(b - a) > 0. \quad (5)$$

Proof. Split every reduced boundary into its *radial* part (circular arcs, normal $\nu = \pm e_r$) and its *angular* part (radial walls, normal $\nu = \pm e_\theta$); all three sets here are polar rectangles, so each perimeter splits exactly, $P(F) = P_r(F) + P_\theta(F)$.

Radial part cancels. Slicing radially, $P_r(F) = \int_{S^1} \sum_{\rho \in \partial^* F_\theta} \rho d\theta$. For a.e. θ the outer crossing of G_θ is $R(\theta)$, the crossings of Q_θ are the inner gap endpoints of E_θ , and together they are exactly the crossings of E_θ with the same radii (the slice balance: $\#\partial^* G_\theta + \#\partial^* Q_\theta = \#\partial^* E_\theta$, weighted by the radii). Hence $P_r(G) + P_r(Q^+) + P_r(Q^-) = P_r(E)$.

Angular part does not. Directly:

$$P_\theta(E) = \underbrace{2(b - a)}_{\text{stem walls } |\theta|=w'/2} + \underbrace{2(R^+ - b)}_{\text{cap walls } |\theta|=w/2} = 2(R^+ - a),$$

$$P_\theta(G) = \underbrace{2(R^+ - a)}_{\text{hull walls } |\theta|=w/2, a < r < R^+}, \quad P_\theta(Q^+) + P_\theta(Q^-) = \underbrace{4(b - a)}_{\text{walls } |\theta|=w'/2 \text{ and } |\theta|=w/2, a < r < b}.$$

Subtracting, $P_\theta(G) + P_\theta(Q^+) + P_\theta(Q^-) - P_\theta(E) = 2(R^+ - a) + 4(b - a) - 2(R^+ - a) = 4(b - a)$, and adding the (vanishing) radial discrepancy gives (??). $\square$

Remark 2 (Diagnosis: the overhang). The culprit is the cap wall $\{|\theta| = w/2, a < r < b\}$, the lateral face of the pocket $Q^\pm$ sitting *underneath* the cap's overhang ($w > w'$). Both sides of this wall — the pocket (empty E) and the exterior (empty E) — have E -density 0, so it lies in neither ∂^*E . Yet it is charged once to $P(G)$ (a radial wall of the hull, where R jumps from R^+ to a) and once to $P(Q^\pm)$. It is double-counted on the left of (??) and absent on the right. This is the exact mechanism by which my earlier “angular part” inequality failed.

Remark 3 (Thin annuli and density do not help). Two natural rescues both fail.

- (i) *Thin annulus.* Take $a = 1$, $b = 1 + \frac{\eta}{2}$, $R^+ = 1 + \eta$, so E lies in the annulus $\{1 < r < 1 + \eta\}$ of width η , while w stays order 1 and $w' \rightarrow 0$. Then the pocket volume is $|Q^+| + |Q^-| \approx (w - w')a(b - a) \sim \eta$, whereas the perimeter deficit is $P(E) - 2\pi \sim \eta$, so $(P(E) - 2\pi)^2 \sim \eta^2 \ll \eta \sim |G \setminus E|$. The quadratic pocket bound is violated by a factor η^{-1} , *inside an arbitrarily thin annulus*.
- (ii) *Two-sided density.* The lollipop is a Lipschitz domain; on ∂E the density is $\approx \frac{1}{2}$ generically and $\approx \frac{1}{4}$ at the finitely many corners, so (1) holds at every boundary point and every scale (down to the corner scale). Density does not see the overhang, because the offending wall is not a boundary point of E at all.

Thus neither the inclusion of (A) nor its density bound is, by itself, enough for the quadratic realization. Something must forbid overhangs.

4 Input (B): realization under no overhang

We isolate the exact condition that makes the decomposition true.

Definition 4 (No overhang). A set of finite perimeter E with $B_{r_-}(0) \subset E \subset B_{r_+}(0)$ has *no overhang about 0* if its radial hull G from (??) satisfies

$$\mathcal{H}^1(\partial^*G \setminus \partial^*E) = 0.$$

Equivalently: $\mathcal{H}^1$ -a.e. point of the hull boundary is a reduced boundary point of E with the same outer normal — the outer profile of E is “self-supporting,” E being full immediately beneath each piece of ∂^*G .

No overhang is implied by, but far weaker than, $C^{1,\gamma}$ star-shapedness; and it is strictly stronger than finite perimeter, the lollipop being the obstruction (there ∂^*G contains the cap-wall segments $\{|\theta| = w/2, a < r < b\}$, which are not in ∂^*E).

Lemma 5 (Slice structure). *For $\mathcal{H}^1$ -a.e. $\theta \in S^1$ the slice E_θ is a one-dimensional set of finite perimeter with $(0, r_-] \subset E_\theta \subset (0, r_+]$, with essential boundary the finite set $\{r_1 < \dots < r_{2m+1}\}$ ($m = m(\theta)$),*

$$E_\theta = (0, r_1) \cup (r_2, r_3) \cup \dots \cup (r_{2m}, r_{2m+1}), \quad R(\theta) = r_{2m+1},$$

$G_\theta = (0, R(\theta))$, and the pocket slice $Q_\theta = (r_1, r_2) \cup \dots \cup (r_{2m-1}, r_{2m})$. Moreover $R \in BV(S^1)$ with values in $[r_-, r_+]$, G is of finite perimeter, and G is a radial graph about 0.

Proof. By Vol’pert’s slicing theorem [1, Thm. 3.108], for $\mathcal{H}^1$ -a.e. θ the slice E_θ is one-dimensional of finite perimeter and $\partial^* E_\theta = (\partial^* E)_\theta$; in dimension one a set of finite perimeter is, up to a null set, a finite union of disjoint open intervals. The inclusion $B_{r_-} \subset E \subset B_{r_+}$ forces $(0, r_-] \subset E_\theta \subset (0, r_+]$, so the first interval starts at 0 and the outermost crossing is $R(\theta) = r_{2m+1}$. The interval structure gives G_θ, Q_θ . That $R \in BV(S^1)$ is the statement that the upper envelope of the BV slice family has bounded variation; G , the polar subgraph of $R \in BV(S^1)$ with values in $[r_-, r_+]$, is then of finite perimeter and is by construction a radial graph. $\square$

Lemma 6 (Perimeter decomposition under no overhang). *If E has no overhang about 0 (Definition ??), with hull G and pockets $\{Q_j\}_{j \in J}$, then, up to $\mathcal{H}^1$ -null sets, $\partial^* G$ and the $\partial^* Q_j$ are pairwise disjoint subsets of $\partial^* E$. Consequently*

$$P(G) + \sum_{j \in J} P(Q_j) \leq P(E). \quad (6)$$

Proof. Write $G = E \sqcup Q \bmod \mathcal{L}^2$ -null, with $Q = \bigsqcup_j Q_j$ and the Q_j the (essentially disjoint, mutually density-0) components of the pocket set.

$\partial^* G \subset \partial^* E$. This is the no-overhang hypothesis.

Each $\partial^* Q_j \subset \partial^* E$. Let $x \in \partial^* Q_j$, so Q_j has density $\frac{1}{2}$ at x . If $x \in \partial^* G$ then $x \in \partial^* E$ by no-overhang, so assume $x \notin \partial^* G$. Since $Q_j \subset G$, density $\frac{1}{2}$ of Q_j at x forces G -density $\geq \frac{1}{2}$, hence (as $x \notin \partial^* G$, where G would have density $\frac{1}{2}$) G has density 1 at x . The other pockets Q_k ($k \neq j$) have density 0 at x (mutually density-0 components). Therefore $E = G \setminus Q$ has density $1 - \frac{1}{2} - 0 = \frac{1}{2}$ at x , with $\nu_E(x) = -\nu_{Q_j}(x)$; thus $x \in \partial^* E$.

Disjointness. If $x \in \partial^* G \cap \partial^* Q_j$ then G and $Q_j \subset G$ both have density $\frac{1}{2}$, so $E \cap B$ has density 0 near x : x separates Q_j (inside G) from $\mathbb{R}^2 \setminus G$ with no E in between, i.e. $x \in \partial^* G \setminus \partial^* E$, a null set by no-overhang. The $\partial^* Q_j$ are mutually $\mathcal{H}^1$ -disjoint as reduced boundaries of mutually density-0 sets.

Conclusion. The sets $\partial^* G, \partial^* Q_1, \partial^* Q_2, \dots$ are, mod $\mathcal{H}^1$ -null, pairwise disjoint subsets of $\partial^* E$, so

$$P(G) + \sum_j P(Q_j) = \mathcal{H}^1\left(\partial^* G \sqcup \bigsqcup_j \partial^* Q_j\right) \leq \mathcal{H}^1(\partial^* E) = P(E). \quad \square$$

Theorem 7 (Radial-graph realization under no overhang). *Let $E \subset \mathbb{R}^2$ have finite perimeter, $|E| = \pi$, $B_{r_-}(0) \subset E \subset B_{r_+}(0)$, the perimeter bound $P(E) \leq 2\pi + \varepsilon$, and no overhang about 0. Let G be the radial hull. Then:*

- (i) G is a radial graph about 0, $G = \{re^{i\theta} : 0 \leq r < R(\theta)\}$ with $R \in BV(S^1)$, $r_- \leq R \leq r_+$;
- (ii) the pocket volume obeys the sharp quadratic bound

$$|G \setminus E| \leq \frac{\varepsilon^2}{4\pi}. \quad (7)$$

Proof. (i) is Lemma 1. For (ii) set $v := |G \setminus E|$, $h := \sum_j P(Q_j)$. By planar isoperimetry applied to G (area $\pi + v$), $P(G) \geq 2\sqrt{\pi(\pi + v)} \geq 2\pi$. Combining with Lemma 2 and $P(E) \leq 2\pi + \varepsilon$,

$$2\pi + \varepsilon \geq P(E) \geq P(G) + h \geq 2\pi + h,$$

so $h \leq \varepsilon$. Apply isoperimetry to each pocket, $|Q_j| \leq P(Q_j)^2/(4\pi)$, and super-additivity $\sum a_j^2 \leq (\sum a_j)^2$ for $a_j \geq 0$:

$$v = \sum_j |Q_j| \leq \frac{1}{4\pi} \sum_j P(Q_j)^2 \leq \frac{1}{4\pi} \left(\sum_j P(Q_j) \right)^2 = \frac{h^2}{4\pi} \leq \frac{\varepsilon^2}{4\pi}. \quad \square$$

Remark 8 (What survives, what does not). The squaring mechanism — isoperimetry per pocket plus super-additivity — never needed regularity, and indeed needs no overhang only through Lemma 2. So the precise content is: *the only place any hypothesis beyond finite perimeter enters realization is the perimeter decomposition, and there the necessary and (essentially) sufficient condition is no-overhang*. The earlier note conflated “regularity-free” with “hypothesis-free”; the lollipop (Proposition ??) shows the two are different. What is genuinely regularity-free is the *passage from no-overhang to the quadratic bound*; what is not free is no-overhang itself.

An unconditional fallback

Without any no-overhang hypothesis one still has, from the inclusion alone, the trivial linear bound

$$|G \setminus E| \leq \pi(\rho_e^2 - \rho_i^2) \leq 2\pi\rho_e \eta, \quad (8)$$

since both E and G lie in the annulus $\{\rho_i < r < \rho_e\}$ outside B_{ρ_i} . This is the bound that holds for the lollipop, and it is the only one available there. The whole point of realization is to upgrade (??) from $O(\eta)$ to $O(\eta^2)$, and Theorem 4 does so exactly under no-overhang.

5 Does input (A) supply no overhang?

This is the crux, and the honest answer is: *partially, above scale r_** . Two-sided density does not by itself remove overhangs (Remark ??(ii)); what removes them is the *tentacle exclusion* that produced density in the first place, applied to the *neck* of an overhang.

Lemma 9 (Necks of overhangs are tentacles). *Suppose Ω satisfies the inclusion (2) and the tentacle exclusion estimate of §2 at scale r_* . Let Q be a pocket of the radial hull whose lateral wall is an overhang, i.e. a component of $\partial^*G \setminus \partial^*\Omega$, and let ℓ be the angular width of Q at its inner radius. If $\ell \geq r_*$ then the cap producing the overhang is attached to $\Omega \setminus B_{\rho_i}$ through a neck of width $\geq cr_*$; more precisely, an overhang with neck width $w' < c_1r_*$ is excluded, because the neck region $\{a < r < b, |\theta - \theta_0| < w'\}$ is an interior tentacle: $|\Omega \cap B_r(x)| \leq Cw'r \ll r^2$ for x at the neck and $r \in [w', r_*]$, forcing $\varepsilon_r^2 \gtrsim r$ and hence Serrin deficit $\gtrsim r_*$.*

Proof sketch. An overhang exists only if the cap (full E at $b < r < R^+$, angular width w) is fed from below through a strictly thinner stem (width $w' < w$): some neck must connect the cap to the bulk $\Omega \setminus B_{\rho_i}$, else the cap is a separate component and contributes a full isoperimetric deficit. At a point x on the neck, for scales $w' \leq r \leq \min(b - a, r_*)$ the slab $\Omega \cap B_r(x)$ is contained in the stem strip of width w' , so $|\Omega \cap B_r(x)| \leq Cw'r \leq C(w'/r)r^2 \ll r^2$ once $w' \ll r$. The tentacle-exclusion estimate $\varepsilon_r^2 \gtrsim |\Omega \cap B_r(x)|^{1/2}$ is then not the sharp form; rather one uses the perimeter lower bound $P(\Omega; B_r(x)) \geq cr$ (two long stem walls of length $\geq r$) against the flux identity to get $\varepsilon_r^2 \gtrsim r$, so any neck with $w' \ll r$ present at a scale $r \leq r_*$ would force $\varepsilon^2 \gtrsim r_*$, contradicting smallness once r is taken $\sim r_*$. Hence necks of width $< c_1r_*$ are excluded above scale r_* . $\square$

Remark 10 (The residual is genuine and sub- r_*). Lemma ?? excludes overhangs whose necks live at scales $\geq r_*$. It says *nothing* about overhangs entirely below scale r_* : a cap of radial extent $\leq \eta$ sitting on a neck of width $< c_1r_*$ at a scale $< r_*$ is invisible to the Serrin deficit. This is the unavoidable η -versus- r_* tension noted before:

- if $\eta \geq r_*$, all annular features are above scale r_* , necks are excluded, Ω has no overhang *modulo a sub- r_* remainder*, and Theorem 4 applies with ε replaced by the above- r_* perimeter deficit;

- if $\eta < r_*$, the entire annular layer is sub- r_* and tentacle exclusion is blind to it; only the trivial bound (??), $|G \setminus \Omega| \lesssim \eta$, survives.

Quantitatively, the overhangs that escape control occupy lateral scale $< r_*$ within a layer of width η , so their total volume is at most $C r_* \cdot (\#\text{necks}) \cdot \eta$; bounding the number of disjoint sub- r_* necks by $P(\Omega)/r_*$ gives the residual pocket bound

$$|G \setminus \Omega| \leq \frac{(P(\Omega) - 2\pi)^2}{4\pi} + C P(\Omega) \eta \quad (\text{above-}r_* \text{ part} + \text{sub-}r_* \text{ residual}), \quad (9)$$

in which the first term is the genuine quadratic gain and the second is the honest residual.

6 Assembly and an honest accounting

Corollary 11 (Selection-free realization, corrected). *Let $\Omega \subset \mathbb{R}^2$, $|\Omega| = \pi$, satisfy input (A): the density (1) and inclusion (2) with width η and $r_- = \rho_i \geq \frac{1}{2}$, at scale r_* . Then Ω has finite perimeter and its radial hull G about z is a radial graph of amplitude $\leq \eta$, with*

$$|G \setminus \Omega| \leq \begin{cases} \frac{(P(\Omega) - 2\pi)^2}{4\pi} + C P(\Omega) \eta, & \eta \geq r_* \text{ (no overhang mod sub-}r_*), \\ 2\pi\rho_e \eta, & \text{always (trivial),} \end{cases}$$

and in either case $\delta_T(G) \leq 2(\delta_T(\Omega) + |G \setminus \Omega|)$. No selection, no $C^{1,\gamma}$, and no atlas-norm smallness is used; only the no-overhang information extracted from η versus r_* .

Proof. Finite perimeter and the inclusion come from (A). When $\eta \geq r_*$, Lemma ?? and Remark ?? give no-overhang modulo a sub- r_* remainder, and Theorem 4 applied to the no-overhang part yields the first bound (??); the trivial bound (??) always holds. The torsion estimate is the planar fill-rescale lemma (monotonicity of torsion under inclusion plus the disk-torsion formula), unchanged from Plan 4. $\square$

Feeding G into the radial-graph theorem (G) then yields $\mathcal{A}_F(\Omega)^2 \lesssim \delta_T(\Omega) + |G \setminus \Omega| + \eta^2$; the realization contributes $|G \setminus \Omega|$ at whichever of the two rates above is in force.

7 What is sharp, what is not, and the remaining gap

Realization is quadratic only under no-overhang. The earlier boxed claim — “realization is sharp; sharpness of the whole route = sharpness of the annulus width η ” — was wrong, because it rested on the false unconditional decomposition (??). The corrected statement is:

$\text{realization is quadratic} \iff \text{no overhang}; \quad \text{finite perimeter alone gives only the linear } O(\eta).$
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The quadratic squaring itself (isoperimetry per pocket plus super-additivity) remains regularity-free; what it now demands as input is the no-overhang condition, not mere finite perimeter.

The rate now has two carriers. The final exponent is governed by *both* the annulus width $\eta = C(\varepsilon^\theta + r_*)$ and the overhang scale r_* :

- if $\eta \geq r_*$, the quadratic term $(P(\Omega) - 2\pi)^2$ dominates and realization gains a power, but a linear residual $C P(\Omega) \eta$ remains from sub- r_* overhangs;
- if $\eta < r_*$, no quadratic gain is available and $|G \setminus \Omega| \lesssim \eta$ is the best one can assert.

At the boundary-layer scales $\varepsilon \sim \delta_T^{1/4}$, $\delta_{\text{iso}} \sim \delta_T^{1/2}$ one has $r_* \sim \varepsilon^2 + \delta_{\text{iso}}^{1/4} \sim \delta_T^{1/8}$ and $\eta \sim \delta_T^{1/8}$, i.e. $\eta \approx r_*$: precisely the borderline, where the residual $CP(\Omega)\eta \sim \delta_T^{1/8}$ and the would-be quadratic term $(P - 2\pi)^2 \sim \delta_T$ are not separated favourably. So at the rate that (A) currently delivers, realization gives no improvement over the trivial $O(\eta)$. An improvement requires (A) to deliver $\eta \gg r_*$, i.e. an annulus width strictly larger than the density scale — the opposite of what one might expect, and the real obstruction.

The remaining gap, restated honestly. The programme trisects:

- **Quadratic squaring:** *done*, regularity-free, given no-overhang (Theorem 4).
- **No-overhang:** reduces to excluding sub- r_* overhangs; tentacle exclusion handles necks at scales $\geq r_*$ (Lemma ??), and the residual (??) is honest and generally nonzero.
- **Inclusion (A):** the Magnanini–Poggesi oscillation bound under density, delivering the annulus width η and the scale r_* ; crucially one needs $\eta \gg r_*$ for the quadratic realization to bite.

Why this matters for the original plan. The original obstruction was phrased as “realization needs $C^{1,\gamma}$, which needs selection.” This note shows the truth is intermediate: realization does *not* need $C^{1,\gamma}$, but it does need no-overhang, which finite perimeter does not give (Proposition ??) and which density gives only above scale r_* . The selection step bought $C^{1,\gamma}$, which trivially implies no-overhang at all scales; the selection-free route must instead win no-overhang from the deficits, and the lollipop pinpoints exactly the configurations it must rule out. That is a sharper and more truthful localisation of the difficulty than the earlier draft, even though it is less triumphant: the realization half is regularity-free, but it is not deficit-free.

References

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